

Matrices and systems of equations

Solving a linear equation for x

We need to find an expression for x

$$y = 3x + z^2x$$

Expand or factorise as needed to separate x from other terms

$$\frac{y}{x} = 3 + z^2$$

Isolate x on one side, expressed in terms of the other variables

$$\frac{1}{x} = \frac{3 + z^2}{y}$$

$$x = \frac{y}{3 + z^2}$$

Two equations, two unknowns

We need to know BOTH variables

$$\begin{aligned}x + 2y &= 12 \\ y - x &= -3\end{aligned}$$

Solve one equation for one unknown

$$y = x - 3$$

Substitute this answer into the other starting equation

$$x + 2(x - 3) = 12$$

Rearrange to solve

$$3x - 6 = 12, \quad x = \frac{18}{3} = 6$$

Substitute this back into the first equation you tinkered with

$$y - 6 = -3, \quad y = 3$$

Systems of equations

Solving systems of equations by substitution works as long as you have one independent equation for every unknown you are trying to find

$$x + y + z = 12,$$

$$2x - y + 3z = 20$$

This system is unsolvable, because we have too few equations

$$x - y - 4 = 0,$$

$$2x - 2y - 8 = 0$$

This system is unsolvable because the two equations are not independent

Large systems

$$2a - 3b + c^2 + \frac{1}{3d} - \sqrt{e} + f^g - 7h - 2i + j^3 + k - 3l = 19$$

I'm going to need 12 equations here...

Matrices

Matrices are like vectors and use many of the same rules (e.g. dot products)

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 6 & 0 & 7 \\ 9 & 1/2 & 0.75 \end{pmatrix} \quad C = \begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix}$$

They can be square or not but are defined by the values in their rows and columns

Matrix notation

$$\begin{matrix} & \begin{matrix} 1 & 2 & n \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ m \end{matrix} & \begin{pmatrix} a_{11} & a_{12} & a_{1n} \\ a_{21} & a_{22} & a_{2n} \\ a_{m1} & a_{m2} & a_{mn} \end{pmatrix} \end{matrix}$$

We number the rows up to m and the columns up to n .

Matrices uses

Phone/PC screen displays, RAM memory on a USB stick, electronics and communication engineering, finding the resonant frequency of bridges...



Addition and subtraction

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} \textit{pizza} & \textit{kebab} \\ \textit{curry} & \textit{soup} \end{pmatrix} = \begin{pmatrix} a + \textit{pizza} & b + \textit{kebab} \\ c + \textit{curry} & d + \textit{soup} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 & -1 \\ 7 & 0 & 4 \end{pmatrix} + \begin{pmatrix} -3 & 4 & 2 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} -1 & 7 & 1 \\ 8 & 0 & 6 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 & -1 \\ 7 & 0 & 4 \end{pmatrix} - \begin{pmatrix} -3 & 4 & 2 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 5 & -1 & -3 \\ 6 & 0 & 2 \end{pmatrix}$$

Multiplication and division by a constant

$$4 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 4 \times 1 & 4 \times 2 \\ 4 \times 3 & 4 \times 4 \end{pmatrix} = \begin{pmatrix} 4 & 8 \\ 12 & 16 \end{pmatrix}$$

$$\frac{1}{3} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{3}{3} & \frac{4}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ 1 & \frac{4}{3} \end{pmatrix}$$

$$\pi \begin{pmatrix} x & y \\ z & \pi \end{pmatrix} = \begin{pmatrix} x\pi & y\pi \\ z\pi & \pi^2 \end{pmatrix}$$

Multiplying two matrices

$$A = \begin{pmatrix} \textcolor{red}{1} & \textcolor{red}{2} \\ \textcolor{green}{3} & \textcolor{green}{4} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} \textcolor{blue}{5} & \textcolor{purple}{6} \\ \textcolor{blue}{7} & \textcolor{purple}{8} \end{pmatrix}$$

$$AB = \begin{pmatrix} \textcolor{red}{1} \times \textcolor{blue}{5} + \textcolor{red}{2} \times \textcolor{blue}{7} & \end{pmatrix} = \begin{pmatrix} 19 & \end{pmatrix}$$

$$AB = \begin{pmatrix} & \textcolor{red}{1} \times \textcolor{purple}{6} + \textcolor{red}{2} \times \textcolor{purple}{8} \end{pmatrix} = \begin{pmatrix} 19 & 22 \end{pmatrix}$$

$$AB = \begin{pmatrix} & \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ \textcolor{green}{3} \times \textcolor{blue}{5} + \textcolor{green}{4} \times \textcolor{blue}{8} & \end{pmatrix}$$

$$AB = \begin{pmatrix} & \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ \textcolor{green}{3} \times \textcolor{purple}{6} + \textcolor{green}{4} \times \textcolor{purple}{8} & 50 \end{pmatrix}$$

Multiplying two matrices of different size

We only require the number of columns in the first matrix to equal the number of rows in the second:

$$A = \begin{pmatrix} 1 & 4 \\ 6 & 3 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

A has two columns and B has two rows, so we can find AB

$$AB = \begin{pmatrix} 1 \times 2 + 4 \times 5 \\ 6 \times 2 + 3 \times 5 \end{pmatrix} = \begin{pmatrix} 22 \\ 27 \end{pmatrix}$$

Give it a try...

If $C = \begin{pmatrix} 7 & 8 \\ 2 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$, find CD

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 \times 5 + 2 \times 7 & \square \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & \square \\ \square & \square \end{pmatrix}$$

$$AB = \begin{pmatrix} \square & 1 \times 6 + 2 \times 8 \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ \square & \square \end{pmatrix}$$

$$AB = \begin{pmatrix} \square & \square \\ 3 \times 5 + 4 \times 8 & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 47 & \square \end{pmatrix}$$

$$AB = \begin{pmatrix} \square & \square \\ \square & 3 \times 6 + 4 \times 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 47 & 50 \end{pmatrix}$$

$$CD = \begin{pmatrix} (7 \times 3) + (8 \times -1) & (7 \times 2) + (8 \times 4) \\ (2 \times 3) + (-1 \times -1) & (2 \times 2) + (-1 \times 4) \end{pmatrix} = \begin{pmatrix} 13 & 46 \\ 7 & 0 \end{pmatrix}$$

Reversing the process

We found that with: $C = \begin{pmatrix} 7 & 8 \\ 2 & -1 \end{pmatrix}$, $D = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$, $CD = \begin{pmatrix} 13 & 46 \\ 7 & 0 \end{pmatrix}$

Now lets calculate CD :

$$DC = \begin{pmatrix} (3 \times 7) + (2 \times 2) & (3 \times 8) + (2 \times -1) \\ (-1 \times 7) + (4 \times 2) & (-1 \times 8) + (4 \times -1) \end{pmatrix} = \begin{pmatrix} 25 & 22 \\ 1 & -12 \end{pmatrix}$$

Matrix multiplication is not commutative!

Special matrices

Diagonal matrix:

Only the diagonal elements take values (though these values can include zero)

$$\begin{pmatrix} 7 & 0 & 0 \\ 0 & -12 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

Identity matrix:

The only elements with any value are the diagonal ones and they are all one

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Identity Matrix

Try multiplying this matrix: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

by the Identity matrix: $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{1 \times 1} + 2 \times 0 & 1 \times 0 + \mathbf{2 \times 1} \\ \mathbf{3 \times 1} + 4 \times 0 & 3 \times 0 + \mathbf{4 \times 1} \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Multiplying a matrix by the identity matrix leaves it unchanged,
like multiplying a scalar quantity by 1

Matrix equations

Try to multiply $A = \begin{pmatrix} 3 & 2 \\ 5 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} x \\ y \end{pmatrix}$

$$AB = \begin{pmatrix} 3x + 2y \\ 5x + 3y \end{pmatrix}$$

This is important, it means we can describe equations using matrices!

Inverse Matrices

We think of the inverse of a function as being its reciprocal:

$$A^{-1} = \frac{1}{A}$$

What we mean by this is that multiplying the inverse of A by A gives us 1:

$$A \left(\frac{1}{A} \right) = 1 \quad \text{or} \quad AA^{-1} = 1$$

But with Matrices, 'one' means the identity matrix, I :

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Finding inverse matrices

Since we know that:

$$A \begin{pmatrix} 1 \\ A \end{pmatrix} = \textcolor{red}{I}$$

It must be true that:

$$\frac{1}{A} = \frac{\textcolor{red}{I}}{A}$$

Fortunately, there is a formula for this:

$$\text{If } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ then } A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Using the inverse method

$$\text{If } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ then } A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

We have:

- 1) Reversed the diagonal elements
- 2) Changed the sign of the other elements
- 3) Multiplied by $\frac{1}{ad-bc}$

Matrices and systems of equations

Consider the simultaneous equations:

$$x + 2y = 13 \qquad 2x - 5y = 8$$

We can write these as matrices:

$$\begin{pmatrix} 1 & 2 \\ 2 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 13 \\ 8 \end{pmatrix}$$

Where: $A = \begin{pmatrix} 1 & 2 \\ 2 & -5 \end{pmatrix}, X = \begin{pmatrix} x \\ y \end{pmatrix}, B = \begin{pmatrix} 13 \\ 8 \end{pmatrix}$

We use A , X and B as the traditional letters but there is nothing special about them – the X here is unrelated to x

Matrix algebra

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -5 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \end{pmatrix}, \quad B = \begin{pmatrix} 13 \\ 8 \end{pmatrix}$$

We have that $AX = B$ and we want to know what the values are in X

If we multiply both sides by A^{-1} we have $A^{-1}AX = A^{-1}B$

But we know that $A^{-1}A = \mathbf{I}$

So we have $\mathbf{I}X = A^{-1}B$, but we know that multiplying a matrix by the identity matrix, $\mathbf{I}$, leaves it unchanged, so...

$$X = A^{-1}B$$

Finding the answers

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -5 \end{pmatrix}$$

$$A^{-1} = \frac{1}{(1)(-5) - (2)(2)} \begin{pmatrix} -5 & -2 \\ -2 & 1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{-9} \begin{pmatrix} -5 & -2 \\ -2 & 1 \end{pmatrix}$$

$$X = A^{-1}B = -\frac{1}{9} \begin{pmatrix} -5 & -2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 13 \\ 8 \end{pmatrix}$$

$$X = -\frac{1}{9} \begin{pmatrix} (-5 \times 13) + (-2 \times 8) \\ (-2 \times 13) + (1 \times 8) \end{pmatrix}$$

$$X = \frac{1}{9} \begin{pmatrix} -81 \\ -18 \end{pmatrix} = \begin{pmatrix} 9 \\ 2 \end{pmatrix}$$

$$x = 9, y = 2$$

And breathe....

I know – it's a lot to take in. It is an entire chapter (27) in the Foundation Maths text.

Solving systems of linear equations is assessed by this module, it is also covered in the ratios and proportions unit.

Matrix use is not assessed by this module, but additional resources are supplied in case you are interested.